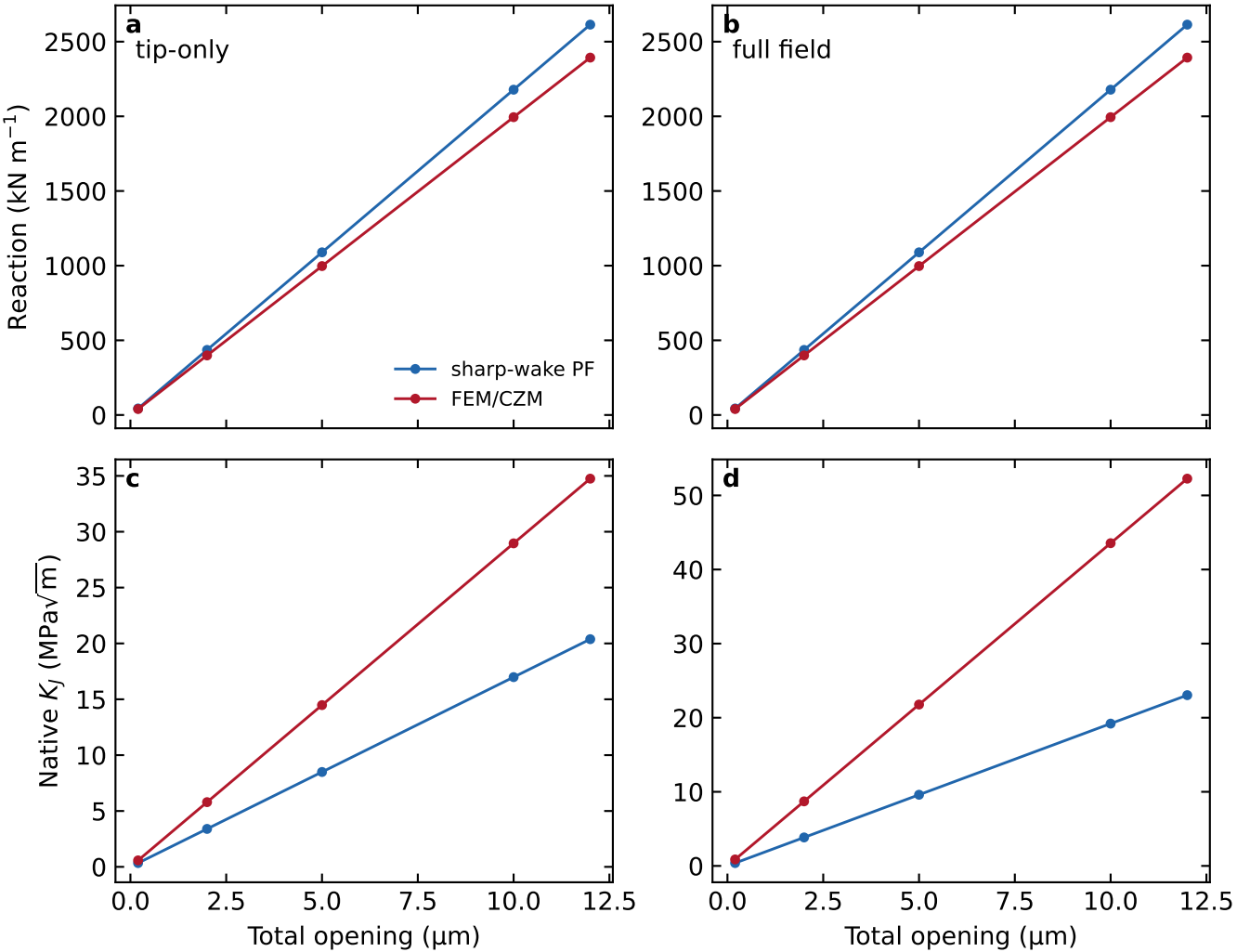
a
tip-only
b
full field
c
d
Reaction (kN m^{-1})
Native K_J (MPa$\sqrt{m}$)
Total opening (μm)
Total opening (μm)
sharp-wake PF
FEM/CZM
0
500
1000
1500
2000
2500
0
5
10
15
20
25
30
35
40
50
0.0
2.5
5.0
7.5
10.0
12.5